EchoProof

Crowdsourced, verifiable environmental sensing rewarded on Solana



The Problem & Our Solution

Environmental data is expensive, centralized, and hard to trust. EchoProof turns smartphones into a verifiable DePIN sensor network on Solana.

- City planners, researchers, and insurers need fresh, granular environmental data but datasets are often stale or unverifiable.
- Contributors use everyday smartphones to passively capture noise, air quality, and GPS-tagged data.
- Each reading is cryptographically signed with device attestation to prove authenticity and resist spoofing.
- Verified contributions are rewarded in SOL/tokens, and buyers pay per query for trusted datasets.

How It Works

A simple flow: capture, prove, reward, query. Solana smart contracts handle verification, incentives, and marketplace access.

- User flow: app passively senses noise/GPS/light, signs reading with device attestation, submits proof.
- Proofs and raw data are stored on IPFS/Arweave; hashes and metadata are anchored on Solana for verification.
- On-chain: an Anchor program checks proof-of-contribution, runs anti-spoofing checks, and triggers token rewards.
- Data buyers query the marketplace by region/time and pay per query via Solana Pay for verified datasets.
- A map dashboard visualizes live data coverage so contributors and buyers see network growth in real time.

Built with

Solana, React Native, Anchor, Rust, IPFS/Arweave, Solana Pay, PostgreSQL for indexing

Why Now & What's Next

DePIN is one of Solana's fastest-growing sectors. Cheap sensors and low-cost transactions make verified crowdsourced data viable now.

- Smartphone sensors are ubiquitous and Solana's low fees make per-contribution micro-rewards economically feasible.
- Next: partner with a city or university for a pilot data purchase to validate real-world demand.
- Next: add hardware sensor support, like air quality dongles, to expand data types beyond phone sensors.
- Next: build reputation-weighted rewards so consistently honest contributors earn more and improve data quality.

Category

DePIN